Database Metadata

Course: DLBDSPBDM01 – Build a Data Mart in SQL; **Programme:** B.Sc. Computer Science; **Author:** Zubair Ahmed Khan; **Matriculation Number:** 92127983; **Tutor:** Prof. Dr. Anna Androvitsanea; **Submission Date:** 2026-07-29

1. Identification

Attribute	Value
Database Name	**AirbnbDB**
Purpose	Coursework data mart modelling an Airbnb-style short-term rental platform, supporting users, guests, hosts, accommodations, bookings, reservations, payments, reviews, complaints, notifications, amenities, locations, and listings.
Version	1.0

2. Technical Environment

Attribute	Value
Database Management System	**MySQL Server 8.0**
Client Tool	**MySQL Workbench 8.0**
SQL Language	**SQL** (DDL and DML)
Storage Engine	InnoDB (default)
Character Set	utf8mb4 (default)
Server Platform	Cross-platform (Windows, macOS, Linux)

3. Schema Statistics

Attribute	Value
Number of Tables	**21**
Number of Entities	**21** (one-to-one with tables)
Number of Relationships	One recursive (Employee–Manager); one ternary (Booking–Payment–Reservation); one many-to-many (Place–Amenity through AmenityListing); plus a network of one-to-many and one-to-one links
Primary Keys	**21** — one per table, integer, named <TableName>ID
Foreign Keys	Implemented across dependent tables; reinforced by ALTER TABLE for late-bound references
Minimum Records per Table	**20**
Total SQL Statements	**70+** (CREATE, ALTER, INSERT, SELECT, SHOW)
Estimated Storage Footprint	Less than 1 MB (small data volume; exact size depends on server configuration and BLOB allocation)

4. Normalisation and Integrity

Attribute	Value
Normalisation	**1NF, 2NF, 3NF** (First, Second, and Third Normal Form)
Primary Key Strategy	Surrogate integer key per table
Foreign Key Strategy	Declarative FOREIGN KEY constraints; orphan rows rejected
Atomicity	All attributes hold single-valued data; no repeating groups
Dependency Hygiene	Non-key attributes depend on the whole primary key, not on other non-key attributes
Referential Integrity	**Enforced** through foreign-key constraints
Domain Integrity	Enforced through data types (DECIMAL(10,2), DATE, DATETIME, VARCHAR, BLOB)

5. Testing

Attribute	Value
Number of Test Cases	**3**
Test Case 1	Retrieval of complete booking information (multi-table join over `Booking`, `Guest`, `User`, `Accommodation`)
Test Case 2	Calculation of host accommodation statistics and income (aggregation over `Accommodation`, `Booking`, `Host`, `User`)
Test Case 3	Retrieval of complete accommodation information (join over `Accommodation`, `Host`, `User`)
Verification Method	Multi-table `JOIN` statements returning non-empty result sets; `SELECT COUNT(*)` per table returning ≥ 20

6. Database Features

Feature	Description
Referential Integrity	Foreign-key constraints reject inserts and updates that would break parent–child links
Dummy Data	At least twenty realistic rows per table, sufficient for non-trivial joins and aggregations
Recursive Relationship	`Employee` references itself through `ManagerID` to model an organisational hierarchy
Ternary Relationship	`Booking`, `Payment`, and `Reservation` form a triple association representing the full reservation lifecycle
Many-to-Many Link	`AmenityListing` resolves the many-to-many between `Place` and `Amenity`
Schema Evolution	`ALTER TABLE` statements add new columns and foreign keys without data loss
Consistency	All non-key attributes depend on the primary key only (3NF); update anomalies are prevented
Scalability	The schema accepts new users, hosts, accommodations, and bookings without structural change
Portability	Pure SQL, no proprietary extensions; runs on any MySQL 8.0 server

7. Repository and Distribution

Attribute	Value
GitHub Repository	https://github.com/zbr-khn/SQL
Submission Folder	`DLBDSPBDM01_Final_Submission/`
SQL File Location	`DLBDSPBDM01_Final_Submission/SQL/airbnb.sql`
Documentation	Abstract, Installation Manual, SQL Documentation, Database Metadata, Submission Checklist
Total Deliverables	Combined PDF portfolio + ZIP folder + individual PDFs

8. Inventory of Tables

#	Table	Purpose
1	`User`	Platform participants (guests and hosts)
2	`Guest`	Specialisation of `User` for booking stays
3	`Host`	Specialisation of `User` for listing accommodations
4	`Accommodation`	Property listed by a host
5	`Booking`	Reservation of an accommodation by a guest
6	`Payment`	Financial settlement of a booking
7	`Reservation`	Confirmed reservation linking booking and payment
8	`Review`	Guest-to-host or host-to-guest review
9	`Complaint`	Issue raised by a guest or host
10	`Notification`	System message directed to a user
11	`Location`	Geographic address (country, region, city, street, post code)
12	`HouseRules`	Set of rules associated with a place
13	`Place`	Physical description of a property (rooms, size, max guests)
14	`Amenity`	Catalogue of amenities (WiFi, AC, pool, etc.)
15	`AmenityListing`	Many-to-many link between `Place` and `Amenity`
16	`Media`	Photos and other media assets
17	`Listing`	Public listing of a place with cover photo and description
18	`Employee`	Back-office staff with a self-referencing manager relationship
19	`Language`	Catalogue of spoken languages

#	Table	Purpose
20	`UserLanguage`	Many-to-many link between `User` and `Language`
21	`Income`	Per-host income record

9. Maintenance Notes

- The script is **not idempotent**; drop the database before re-running.
- The `ALTER TABLE` block in Section D is essential; without it, the `PlaceID` foreign key in `Accommodation` is not created.
- The `PaymentDate` column is added after the `Payment` inserts through a separate `ALTER TABLE`, in order to preserve the linear execution order of the script.
- For very large datasets, indexes on `Booking.GuestID`, `Booking.AccommodationID`, `Accommodation.HostID`, `Reservation.BookingID`, and `Reservation.PaymentID` would accelerate the test-case joins.